MLOPS

MLOps Visual Guide

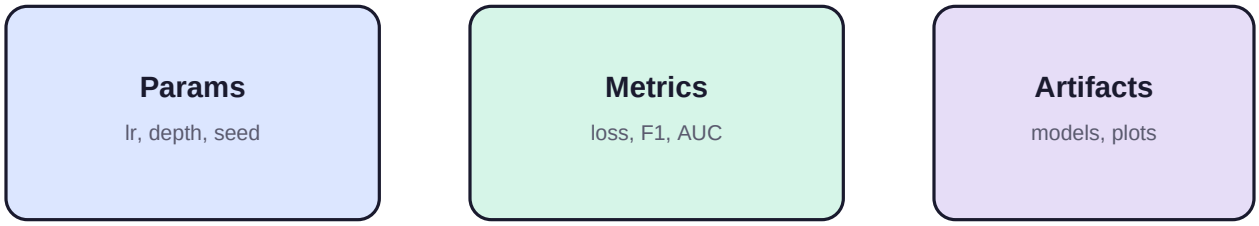
Operating ML in production:
tracking, registry, CI/CD, feature stores.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

Experiment Tracking

Every run recorded, comparable, reproducible.



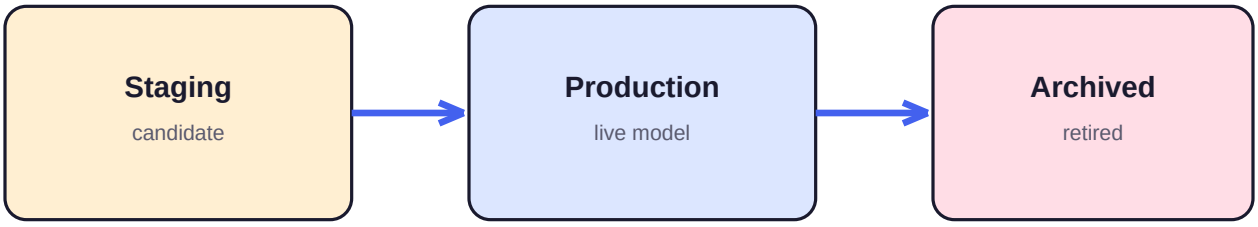
MLflow / Weights & Biases: log everything per run. 'Best model' must be a query, not a memory.

Log the git hash, data version, and environment with every run — otherwise results aren't reproducible.

STEP 2/6

The Model Registry

From experiment to governed artifact.

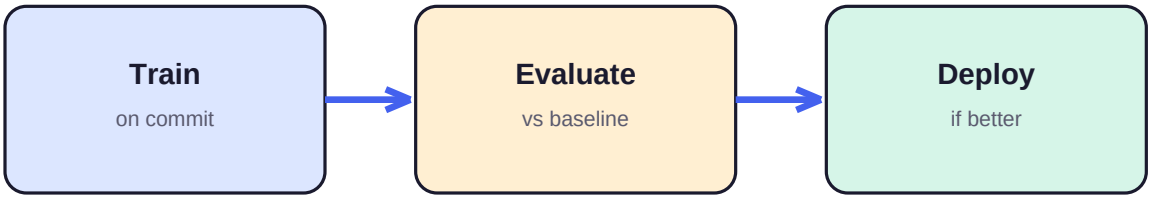


Promotion is a gated decision: champion/challenger tests, sign-offs, and automatic rollback on regression.

STEP 3/6

CI/CD for ML

Test the pipeline, not just the code.



Unit-test data schemas and feature code; integration-test the training pipeline on a data sample.

ML CI is slower than software CI — cache datasets and use small representative samples for fast feedback.

STEP 4/6

Feature Stores

One definition of each feature, everywhere.

Offline store

training: point-in-time-correct joins for training sets.

Online store

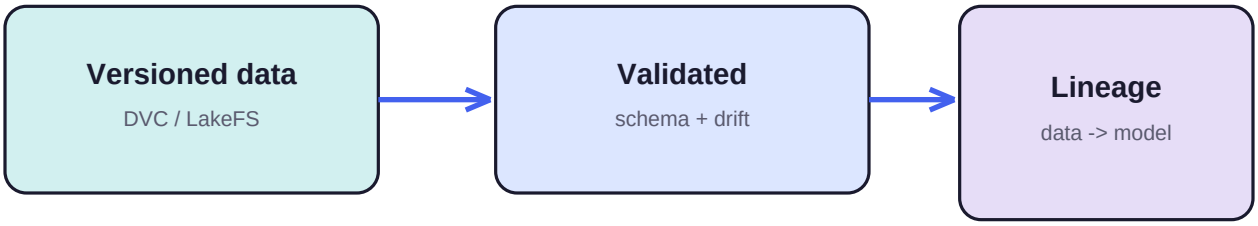
serving: low-latency lookups for live predictions.

Training/serving skew — features computed differently in prod — is a top silent model killer. Feature stores prevent it.

STEP 5/6

Data in the Pipeline

Models are downstream of data.

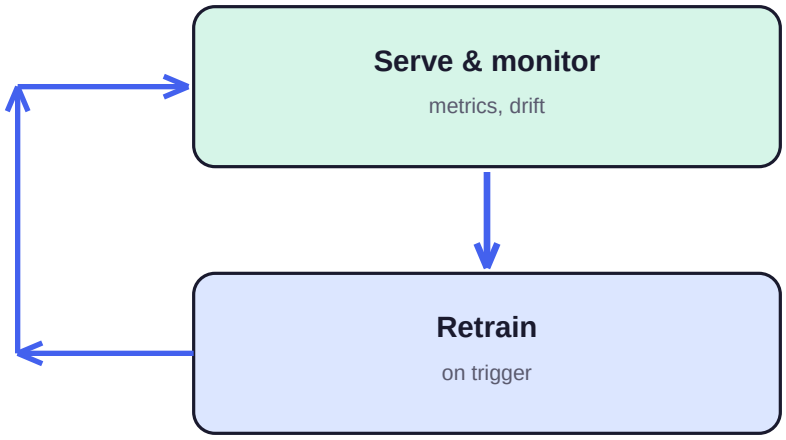


Every training run must record exactly which data snapshot it used — 'the latest CSV' is not a version.

STEP 6/6

The MLOps Loop

Monitor, trigger, retrain, redeploy.



Retraining triggers: schedule, drift detection, or performance decay — never 'vibes'.

Close the loop with automated rollback: a worse challenger must never silently become champion.

STEP ✓I✓

Key Takeaways

MLOps — Remember

- Track every experiment: params, metrics, artifacts, git hash, data version.
- The registry governs promotion: staging -> production -> archived, with gates.
- CI/CD for ML tests data and pipelines, not just code.
- Feature stores kill training/serving skew with one feature definition.
- Monitor in prod; retrain on triggers; roll back automatically on regression.

Keep exploring

Pair this guide with the matching video walkthroughs
and hands-on labs at interstitiumlabs.dev